

Structural-Break-First Statistical Framework

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Abstract

This paper develops a structural-break-first statistical framework in which regime instability is treated as primitive rather than exceptional. Classical inference typically begins with a stable data-generating process and subsequently tests for violations, disturbances, or breaks. The framework developed here reverses that order: the primary inferential object is the regime structure that generated the data. Parameters, causal effects, forecasts, risks, and policy conclusions are then defined locally within regimes and aggregated only under explicit structural compatibility conditions. The framework integrates statistics, econometrics, finance, political science, geopolitics, international relations, warfare economics, data science, and machine learning. It introduces regime-local likelihoods, structural variance decomposition, a Structural Break Index, break-risk forecasting, causal-break inference, and algorithms for segmentation, classification, and prediction. Applications are developed for macroeconomic crises, financial crashes, sanctions, alliance shifts, military campaigns, institutional reform, and conflict systems. The central doctrine is simple: no global inference without structural stability.

The paper ends with “The End”

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1 Introduction

Many statistical models begin with an implicit stability premise. A sequence of observations

$$X_1, \dots, X_T$$

is often treated as arising from a common data-generating process

$$X_t \sim P_\theta,$$

possibly with stochastic noise, dependence, or covariate adjustment. Structural change is then treated as a complication: a break, shock, intervention, discontinuity, crisis, or outlier.

The structural-break-first statistical framework reverses this order. It begins from the premise that many real systems are not globally stable. They are locally stable within regimes and structurally discontinuous across regimes. Thus the primitive object is not merely a parameter θ , but a regime-break structure:

$$\mathcal{S} = (K, \{\tau_k\}_{k=0}^K, \{\theta_k\}_{k=1}^K),$$

where K is the number of regimes, τ_k are break dates, and θ_k are regime-specific parameters.

The central principle is:

Estimate the regime structure before estimating the parameter.

Equivalently:

No global inference without structural stability.

This principle matters in domains where statistical objects are altered by crises, institutions, strategic behavior, violence, market panic, legal reforms, technological discontinuities, and geopolitical realignments. A single coefficient or global average may be misleading when the meaning of the data itself changes across regimes.

2 Motivation

Structural breaks appear in many fields:

- in economics, through recessions, inflation regimes, reforms, debt crises, and development transitions;
- in finance, through crashes, liquidity freezes, volatility explosions, and contagion;
- in political science, through regime changes, coups, electoral shocks, constitutional reform, and state failure;
- in geopolitics and international relations, through wars, sanctions, alliance shifts, and balance-of-power transitions;

- in warfare, through mobilization, escalation, attrition, tactical adaptation, and logistical collapse;
- in data science and AI, through concept drift, distribution shift, adversarial manipulation, and model decay.

The classical question is often:

What is the value of θ ?

The structural-break-first question is:

Which regimes generated the data, and what is true inside each regime?

3 Primitive Objects

Let

$$\mathcal{D}_T = \{X_t\}_{t=1}^T$$

denote the observed data sequence, where each observation satisfies

$$X_t \in \mathcal{X}.$$

The space $\mathcal{X}$ may contain scalar observations, vector observations, networks, texts, images, financial panels, military event data, or institutional records.

Definition 1 (Regime). *A regime is a locally stable statistical environment*

$$R_k = (I_k, P_k, \theta_k),$$

where

$$I_k = \{\tau_{k-1} + 1, \dots, \tau_k\}$$

is the interval of regime k , P_k is the regime-specific probability law, and θ_k is the regime-specific parameter.

Definition 2 (Structural Break). *A structural break is a time τ_k such that*

$$P_{\theta_k} \neq P_{\theta_{k+1}}.$$

The break may affect means, variances, dependence, causal mechanisms, measurement systems, institutions, or strategic behavior.

Definition 3 (Break Field). *The break field is the set of structural discontinuities:*

$$\mathcal{B} = \{\tau_1, \dots, \tau_{K-1}\}.$$

Definition 4 (Regime Map). *The regime map is the function*

$$\rho : \{1, \dots, T\} \rightarrow \{1, \dots, K\}$$

defined by

$$\rho(t) = k \iff \tau_{k-1} < t \leq \tau_k.$$

Thus

$$X_t \sim P_{\theta_{\rho(t)}}.$$

4 Axioms

Axiom 1 (Local Stability). *For each regime R_k , there exists a parameter θ_k such that observations inside the regime are governed by a stable law:*

$$X_t \mid \rho(t) = k \sim P_{\theta_k}.$$

Stability is local, not global.

Axiom 2 (Structural Discontinuity). *There exist times at which the governing law changes:*

$$P_{\theta_k} \neq P_{\theta_{k+1}}.$$

The stationary model is the special case $K = 1$.

Axiom 3 (Break Priority). *Inference must first account for structural breaks before estimating global quantities. The primary estimator is*

$$\hat{\mathcal{S}} = \left(\hat{K}, \{\hat{\tau}_k\}_{k=0}^{\hat{K}}, \{\hat{\theta}_k\}_{k=1}^{\hat{K}} \right).$$

Axiom 4 (Regime Non-Equivalence). *Parameters from different regimes are not automatically comparable. The expression*

$$\theta_i - \theta_j$$

is meaningful only when θ_i and θ_j have the same semantic interpretation across regimes.

Axiom 5 (Break Endogeneity). *Breaks may be caused by variables inside the system:*

$$\mathbb{P}(\tau_k = t) = h(X_1, \dots, X_{t-1}, Z_t),$$

where Z_t denotes exogenous shocks, policy actions, institutional changes, wars, sanctions, or strategic interventions.

5 Likelihood Formulation

Suppose regime k has density

$$f(x_t; \theta_k).$$

Given a structural object

$$\mathcal{S} = (K, \{\tau_k\}_{k=0}^K, \{\theta_k\}_{k=1}^K),$$

the likelihood is

$$L(\mathcal{S}; \mathcal{D}_T) = \prod_{k=1}^K \prod_{t=\tau_{k-1}+1}^{\tau_k} f(x_t; \theta_k).$$

The log-likelihood is

$$\ell(\mathcal{S}) = \sum_{k=1}^K \sum_{t=\tau_{k-1}+1}^{\tau_k} \log f(x_t; \theta_k).$$

A penalized estimator is

$$\hat{\mathcal{S}} = \arg \max_{\mathcal{S}} [\ell(\mathcal{S}) - \lambda \Phi(K, T)],$$

where $\Phi(K, T)$ penalizes over-fragmentation. Common choices include

$$\Phi(K, T) = K,$$

$$\Phi(K, T) = K \log T,$$

and

$$\Phi(K, T) = \sum_{k=1}^K \dim(\theta_k) \log T.$$

6 Break Types

Structural breaks are not limited to changes in means. Table 1 gives a taxonomy.

Table 1: Types of structural breaks

Break type	Mathematical form	Interpretation
Mean break	$\mu_k \neq \mu_{k+1}$	Level shift
Variance break	$\sigma_k^2 \neq \sigma_{k+1}^2$	Volatility shift
Covariance break	$\Sigma_k \neq \Sigma_{k+1}$	Correlation shift
Tail break	$\alpha_k \neq \alpha_{k+1}$	Extreme-event shift
Dependence break	$P(X_t X_{t-1})$ changes	Dynamic-law shift
Causal break	$G_k \neq G_{k+1}$	Causal-graph shift
Measurement break	$M_k \neq M_{k+1}$	Observation-process shift
Institutional break	$\mathcal{I}_k \neq \mathcal{I}_{k+1}$	Rule-system shift
Strategic break	$A_k \neq A_{k+1}$	Agent-behavior shift
Ontological break	$\mathcal{X}_k \neq \mathcal{X}_{k+1}$	Meaning/support shift

7 Regime Map and Visualization

Figure 1 represents the core idea graphically.

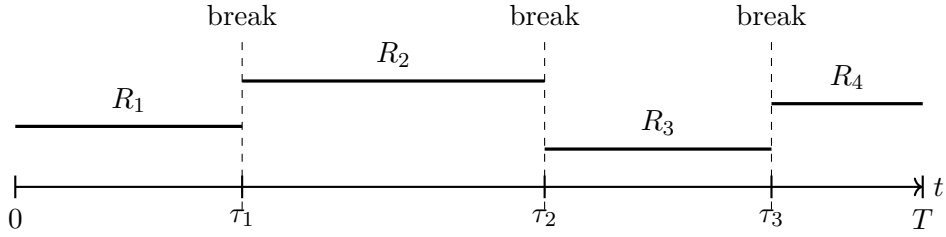


Figure 1: A structural-break-first regime map. Each interval is locally stable, but the full sample is not assumed globally stable.

8 Regime-Local Statistics

Given estimated regimes, define local statistics.

The regime-local mean is

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{t=\hat{\tau}_{k-1}+1}^{\hat{\tau}_k} X_t,$$

where

$$n_k = \hat{\tau}_k - \hat{\tau}_{k-1}.$$

The regime-local variance is

$$\hat{\sigma}_k^2 = \frac{1}{n_k - 1} \sum_{t=\hat{\tau}_{k-1}+1}^{\hat{\tau}_k} (X_t - \hat{\mu}_k)^2.$$

For a generic loss function L , the regime-local empirical risk is

$$\hat{\mathcal{R}}_k = \frac{1}{n_k} \sum_{t=\hat{\tau}_{k-1}+1}^{\hat{\tau}_k} L(X_t, \hat{\theta}_k).$$

For causal inference, if D_t is treatment and Y_t is outcome, the regime-local average treatment effect is

$$\Delta_k = \mathbb{E}[Y_t(1) - Y_t(0) \mid \rho(t) = k].$$

9 Global Quantities as Derived Objects

In this framework, global quantities are not primitive. They are regime-weighted constructions.

Let

$$w_k = \frac{n_k}{T}.$$

The global mean is

$$\bar{\mu} = \sum_{k=1}^K w_k \mu_k.$$

The global variance decomposes into within-regime and between-regime components:

$$\text{Var}(X) = \sum_{k=1}^K w_k \sigma_k^2 + \sum_{k=1}^K w_k (\mu_k - \bar{\mu})^2.$$

The second term is the structural dispersion:

$$SD = \sum_{k=1}^K w_k (\mu_k - \bar{\mu})^2.$$

Theorem 1 (Structural Break Decomposition Theorem). *Let X be generated by K regimes with regime means μ_k , variances σ_k^2 , and weights w_k . Then*

$$\text{Var}(X) = \underbrace{\sum_{k=1}^K w_k \sigma_k^2}_{\text{within-regime variance}} + \underbrace{\sum_{k=1}^K w_k (\mu_k - \bar{\mu})^2}_{\text{between-regime structural variance}}.$$

Proof. Let R denote the regime random variable, with

$$\mathbb{P}(R = k) = w_k.$$

By the law of total variance,

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid R)] + \text{Var}(\mathbb{E}[X \mid R]).$$

Since

$$\text{Var}(X \mid R = k) = \sigma_k^2,$$

we have

$$\mathbb{E}[\text{Var}(X \mid R)] = \sum_{k=1}^K w_k \sigma_k^2.$$

Also,

$$\mathbb{E}[X \mid R = k] = \mu_k,$$

so

$$\text{Var}(\mathbb{E}[X \mid R]) = \sum_{k=1}^K w_k (\mu_k - \bar{\mu})^2.$$

Therefore,

$$\text{Var}(X) = \sum_{k=1}^K w_k \sigma_k^2 + \sum_{k=1}^K w_k (\mu_k - \bar{\mu})^2.$$

□

10 Structural Break Index

Define the Structural Break Index by

$$SBI = \frac{\sum_{k=1}^K w_k (\mu_k - \bar{\mu})^2}{\text{Var}(X)}.$$

Thus SBI measures the share of total variance attributable to structural differences across regimes.

Table 2: Interpretation of the Structural Break Index

Range	Interpretation
$0 \leq SBI < 0.2$	Weak structural instability
$0.2 \leq SBI < 0.5$	Moderate structural instability
$0.5 \leq SBI < 0.8$	Strong structural instability
$0.8 \leq SBI \leq 1$	Regime-dominated system

11 Break Detection

For a candidate break at time τ , compare the one-regime model to the two-regime model.

The no-break likelihood is

$$L_0 = \prod_{t=1}^T f(X_t; \hat{\theta}).$$

The break likelihood is

$$L_1(\tau) = \prod_{t=1}^{\tau} f(X_t; \hat{\theta}_1) \prod_{t=\tau+1}^T f(X_t; \hat{\theta}_2).$$

The break gain is

$$G(\tau) = 2[\ell_1(\tau) - \ell_0].$$

A break is accepted when

$$G(\tau) > c_T,$$

where c_T is a complexity-adjusted threshold. A simple penalty rule is

$$G(\tau) > p \log T,$$

where p is the number of additional parameters introduced by the break.

12 Multiple-Break Estimation

For multiple breaks, estimate

$$\hat{\mathcal{B}} = \arg \max_{\mathcal{B}} \left[\sum_{k=1}^K \ell_k(\hat{\theta}_k) - \lambda K \log T \right].$$

Equivalently,

$$\hat{\mathcal{B}} = \arg \min_{\mathcal{B}} \left[\sum_{k=1}^K C(X_{\tau_{k-1}+1:\tau_k}) + \lambda K \log T \right],$$

where $C(\cdot)$ is a segment cost.

For Gaussian mean breaks,

$$C(I_k) = \sum_{t \in I_k} (X_t - \bar{X}_k)^2.$$

For likelihood models,

$$C(I_k) = -\max_{\theta_k} \sum_{t \in I_k} \log f(X_t; \theta_k).$$

For machine-learning models,

$$C(I_k) = \sum_{t \in I_k} L(Y_t, \hat{f}_k(X_t)).$$

13 Algorithmic Framework

Algorithm 1 (Structural-Break-First Inference). *Given data $\mathcal{D}_T$, model class $\mathcal{M}$, penalty λ , and minimum regime length m :*

1. *Generate candidate break sets*

$$\mathcal{B} = \{\tau_1, \dots, \tau_{K-1}\}$$

satisfying

$$\tau_k - \tau_{k-1} \geq m.$$

2. *For each segment I_k , estimate*

$$\hat{\theta}_k = \arg \max_{\theta_k} \sum_{t \in I_k} \log f(X_t; \theta_k).$$

3. *Select the structural object*

$$\hat{\mathcal{S}} = \arg \max_{\mathcal{S}} [\ell(\mathcal{S}) - \lambda K \log T].$$

4. *Classify each break by type:*

$$\Delta\mu, \quad \Delta\sigma^2, \quad \Delta\Sigma, \quad \Delta G, \quad \Delta M, \quad \Delta\mathcal{I}.$$

5. *Estimate regime-local quantities:*

$$\hat{\theta}_1, \dots, \hat{\theta}_K.$$

6. *Compute diagnostics:*

$$SBI, \quad n_k, \quad \hat{\sigma}_k^2, \quad G(\hat{\tau}_k).$$

7. *Report the regime map rather than only a global coefficient.*

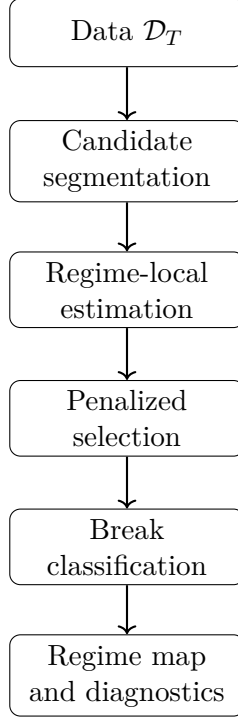


Figure 2: Structural-break-first inference pipeline.

14 Break Uncertainty

A break date should not be reported only as a point estimate. Define a confidence set:

$$C_\alpha(\tau) = \{t : G(\hat{\tau}) - G(t) \leq c_\alpha\}.$$

This gives a break interval:

$$\hat{\tau} \in [\tau_L, \tau_U].$$

In crises, recessions, wars, reforms, and crashes, the statistically plausible break interval may be more meaningful than a single date.

15 Forecasting Under Break Risk

Classical forecasting often uses

$$\hat{Y}_{T+h} = f(X_T; \hat{\theta}).$$

Structural-break-first forecasting conditions on both the current regime and future break risk.

Let

$$\pi_h = \mathbb{P}(\text{break occurs between } T \text{ and } T+h \mid \mathcal{D}_T).$$

Then

$$\mathbb{E}[Y_{T+h} \mid \mathcal{D}_T] = (1 - \pi_h)\mathbb{E}[Y_{T+h} \mid R_T] + \pi_h\mathbb{E}[Y_{T+h} \mid R_{\text{new}}].$$

Forecast risk decomposes as

$$\text{Risk}_{\text{forecast}} = \text{Risk}_{\text{within}} + \text{Risk}_{\text{break}}.$$

Define horizon- h break risk:

$$BR_h = \mathbb{P}(\tau_{\text{next}} \leq T+h \mid \mathcal{D}_T).$$

Table 3: Volatility and break risk

System type	Volatility	Break risk	Interpretation
Stable-noisy	High	Low	Fluctuates but survives
Quiet-fragile	Low	High	Calm before rupture
Turbulent-fragile	High	High	Crisis-prone
Stable-calm	Low	Low	Robust stability

16 Structural-Break-First Regression

Consider the classical regression

$$Y_t = X_t^\top \beta + \varepsilon_t.$$

The structural-break-first version is

$$Y_t = X_t^\top \beta_{\rho(t)} + \varepsilon_t,$$

where

$$\varepsilon_t \sim F_{\rho(t)}.$$

The regression object is

$$\mathcal{R} = (K, \{\tau_k\}, \{\beta_k\}, \{F_k\}).$$

The global regression coefficient is valid only under

$$\beta_1 = \beta_2 = \dots = \beta_K.$$

Proposition 1 (Invalidity of Global Regression Under Structural Instability). *Suppose*

$$Y_t = X_t^\top \beta_{\rho(t)} + \varepsilon_t$$

with at least two regimes satisfying $\beta_i \neq \beta_j$. Then the pooled coefficient generally estimates a mixture object rather than any regime-local structural parameter.

Proof. The pooled least-squares estimator solves

$$\hat{\beta} = \left(\sum_{t=1}^T X_t X_t^\top \right)^{-1} \left(\sum_{t=1}^T X_t Y_t \right).$$

Substituting the regime-local model gives

$$\hat{\beta} = \left(\sum_{t=1}^T X_t X_t^\top \right)^{-1} \left(\sum_{k=1}^K \sum_{t \in I_k} X_t X_t^\top \beta_k + \sum_{t=1}^T X_t \varepsilon_t \right).$$

Ignoring estimation noise, the estimand is

$$\beta^* = \left(\sum_{k=1}^K Q_k \right)^{-1} \left(\sum_{k=1}^K Q_k \beta_k \right),$$

where

$$Q_k = \sum_{t \in I_k} X_t X_t^\top.$$

Unless all β_k are equal or special weighting conditions hold, β^* is a weighted mixture, not a structural parameter of any single regime. $\square$

17 Structural-Break-First Causal Inference

Let $Y_t(d)$ denote the potential outcome under treatment d . The classical average treatment effect is

$$ATE = \mathbb{E}[Y(1) - Y(0)].$$

The regime-local treatment effect is

$$ATE_k = \mathbb{E}[Y_t(1) - Y_t(0) \mid \rho(t) = k].$$

The global treatment effect is

$$ATE = \sum_{k=1}^K w_k ATE_k,$$

but this aggregate is meaningful only under causal compatibility across regimes.

Define causal instability:

$$CI = \sum_{k=1}^K w_k (ATE_k - ATE)^2.$$

High CI means treatment effects are structurally unstable.

18 Causal-Break Graphs

A causal break may alter the causal graph itself. Let

$$G_k = (V, E_k)$$

be the causal graph in regime k . A causal break occurs when

$$E_k \neq E_{k+1}.$$

Figure 3 illustrates a transition from a peace economy to a war economy.

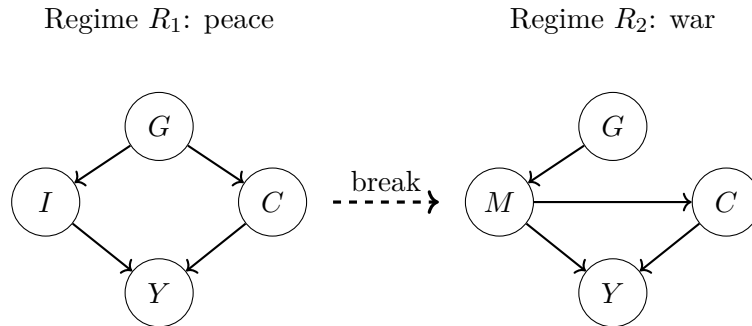


Figure 3: A causal break: the graph governing output changes from a peace economy to a war economy. Here G denotes government, I investment, C consumption, M mobilization, and Y output.

19 Economics

In macroeconomics, structural breaks arise through monetary-policy shifts, fiscal crises, commodity shocks, reforms, pandemics, wars, debt defaults, and institutional transitions.

Let output growth be

$$g_t = \alpha_{\rho(t)} + \beta_{\rho(t)} r_t + \gamma_{\rho(t)} \pi_t + \varepsilon_t,$$

where r_t is the interest rate and π_t is inflation. A structural break occurs if

$$(\alpha_k, \beta_k, \gamma_k) \neq (\alpha_{k+1}, \beta_{k+1}, \gamma_{k+1}).$$

The Keynesian multiplier may be regime-specific:

$$m_k = \frac{1}{1 - c_k(1 - \tau_k) + q_k},$$

where c_k is the marginal propensity to consume, τ_k is the tax rate, and q_k is the import leakage parameter. A pooled multiplier

$$m$$

may be meaningless during a transition from normalcy to crisis.

Table 4: Economic interpretation of regimes

Regime	Statistical signature	Economic interpretation
Expansion	$\mu_g > 0$, σ_g^2 moderate	Growth regime
Recession	$\mu_g < 0$	Demand contraction
Stagflation	$\mu_g < 0$, $\mu_\pi > 0$	Supply-side crisis
Debt crisis	rising spreads, fiscal stress	Sovereign-risk regime
Reform transition	parameter instability	Institutional restructuring
War economy	output measurement shift	Mobilization regime

20 Finance

In finance, structural breaks appear through crashes, bubbles, liquidity freezes, volatility regimes, contagion, regulatory shifts, and monetary-policy surprises.

Let asset returns satisfy

$$r_t = \mu_{\rho(t)} + \sigma_{\rho(t)} \varepsilon_t.$$

A volatility break occurs when

$$\sigma_k^2 \neq \sigma_{k+1}^2.$$

For a portfolio with weights w , the regime-local variance is

$$\text{Var}_k(r_p) = w^\top \Sigma_k w.$$

Regime-local Value-at-Risk at confidence level q is

$$\text{VaR}_{q,k} = -\inf\{x : F_k(x) \geq q\}.$$

A structural-break-first risk system reports

$$\{\text{VaR}_{q,1}, \dots, \text{VaR}_{q,K}\}$$

rather than a single historical VaR .

Proposition 2 (Crash Underestimation by Pooled Volatility). *Suppose returns have two regimes with variances $\sigma_1^2 < \sigma_2^2$ and the current regime is R_2 . A pooled variance estimator*

$$\hat{\sigma}^2 = w_1\sigma_1^2 + w_2\sigma_2^2$$

underestimates current variance whenever $w_1 > 0$.

Proof. Since $\sigma_1^2 < \sigma_2^2$ and $w_1 > 0$ with $w_1 + w_2 = 1$,

$$\hat{\sigma}^2 = w_1\sigma_1^2 + w_2\sigma_2^2 < w_1\sigma_2^2 + w_2\sigma_2^2 = \sigma_2^2.$$

Thus pooled variance underestimates current crash-regime variance. □

21 Political Science

Political systems often shift across electoral regimes, constitutional regimes, repression regimes, protest regimes, and state-capacity regimes.

Let Y_t denote government stability, P_t protest intensity, E_t economic stress, and S_t security-force loyalty. A regime-local instability equation may be

$$Y_t = \alpha_{\rho(t)} + \beta_{\rho(t)}P_t + \gamma_{\rho(t)}E_t + \delta_{\rho(t)}S_t + \varepsilon_t.$$

A political break occurs when

$$(\beta_k, \gamma_k, \delta_k) \neq (\beta_{k+1}, \gamma_{k+1}, \delta_{k+1}).$$

Thus the effect of protest on regime stability may be weak in one institutional regime and strong in another.

22 Geopolitics and International Relations

In international relations, systems may shift from unipolarity to multipolarity, deterrence to escalation, cooperation to sanctions, or stable alliances to fragmented blocs.

Let $A_{ij,t}$ denote alliance strength between states i and j . A geopolitical regime can be represented by a network

$$G_t = (V, E_t, W_t),$$

where $W_t = (A_{ij,t})$.

A network structural break occurs when

$$W_k \neq W_{k+1}.$$

Balance-of-power stress may be written as

$$B_t = \sum_{i=1}^N \left(s_{i,t} - \frac{1}{N} \right)^2,$$

where $s_{i,t}$ is the share of total system capability held by state i .

A break in the international system occurs when

$$P(B_t \mid R_k) \neq P(B_t \mid R_{k+1}).$$

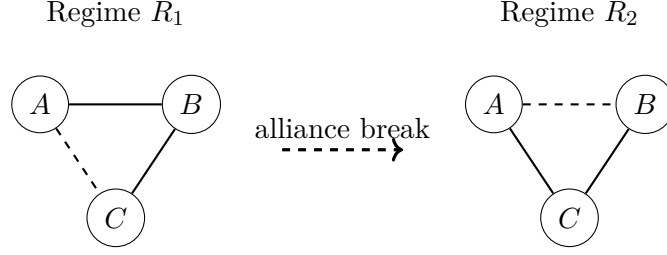


Figure 4: A geopolitical network break. Solid lines denote strong alignment; dashed lines denote weak alignment.

23 Warfare Economics

War economies are structurally distinct from peace economies. Mobilization changes production, labor allocation, fiscal capacity, trade patterns, logistics, inflation, and measurement.

Let civilian output be Y_t^c , military output be Y_t^m , and total measured output be

$$Y_t = Y_t^c + Y_t^m.$$

A mobilization regime may satisfy

$$Y_t^m = \phi_{\rho(t)} G_t,$$

where G_t is government military expenditure and $\phi_{\rho(t)}$ is regime-local military production efficiency.

The war-economy resource constraint is

$$C_t + I_t + M_t + L_t = Y_t + B_t,$$

where C_t is consumption, I_t investment, M_t military absorption, L_t logistical loss, and B_t borrowing or external support.

A warfare economic break occurs when

$$\phi_k \neq \phi_{k+1},$$

or when the resource constraint itself changes due to blockade, sanctions, mobilization, occupation, or destruction of infrastructure.

24 Warfare and Conflict Systems

Military campaigns exhibit structural breaks through escalation, tactical adaptation, attrition thresholds, logistical exhaustion, morale collapse, command disruption, and technological surprise.

Let F_t denote fighting capacity:

$$F_t = a_{\rho(t)} T_t + b_{\rho(t)} L_t + c_{\rho(t)} C_t + d_{\rho(t)} M_t + \varepsilon_t,$$

where T_t is troop strength, L_t logistics, C_t command coherence, and M_t morale.

A military break occurs when

$$(a_k, b_k, c_k, d_k) \neq (a_{k+1}, b_{k+1}, c_{k+1}, d_{k+1}).$$

Attrition may be modeled as

$$F_{t+1} = F_t - \alpha_{\rho(t)} A_t + \eta_t,$$

where A_t is adversary pressure and $\alpha_{\rho(t)}$ is regime-specific attrition sensitivity.

Table 5: Conflict-regime break types

Break type	Statistical signature	Conflict interpretation
Escalation break	jump in intensity	wider battlefield or higher force use
Attrition break	change in loss rate	changed survivability or exposure
Logistics break	supply covariance shift	supply-chain disruption
Morale break	performance collapse	cohesion or legitimacy failure
Technology break	effect-size discontinuity	new weapon or countermeasure
Command break	dependence disruption	coordination failure
Measurement break	reporting-process change	censorship or fog of war

25 Data Science

In data science, structural breaks correspond to concept drift, covariate shift, label shift, target drift, platform changes, and adversarial perturbation.

Let supervised learning data be

$$\{(X_t, Y_t)\}_{t=1}^T.$$

A structural-break-first prediction model is

$$Y_t = f_{\rho(t)}(X_t) + \varepsilon_t.$$

The learning object is

$$(\hat{\rho}, \hat{f}_1, \dots, \hat{f}_K).$$

A structural-break-first empirical risk is

$$\mathcal{L} = \sum_{t=1}^T L(Y_t, \hat{f}_{\hat{\rho}(t)}(X_t)) + \lambda \hat{K}.$$

Algorithm 2 (Structural-Break-First Machine Learning). *Given labeled data $\{(X_t, Y_t)\}_{t=1}^T$:*

1. *Estimate candidate break field $\hat{\mathcal{B}}$ using distributional, predictive, or representation-based divergence.*
2. *Assign each observation to a regime:*

$$\hat{\rho}(t) = k.$$

3. *Train local predictors:*

$$\hat{f}_k = \arg \min_{f \in \mathcal{F}} \sum_{t: \hat{\rho}(t)=k} L(Y_t, f(X_t)).$$

4. *Train an online regime classifier:*

$$\hat{g}_t = g(X_1, \dots, X_t).$$

5. *Predict by*

$$\hat{Y}_{t+1} = \hat{f}_{\hat{g}_t}(X_{t+1}).$$

6. *Monitor residuals for new breaks:*

$$e_t = Y_t - \hat{f}_{\hat{g}_{t-1}}(X_t).$$

26 AI and Machine Learning Diagnostics

Let $Z_t = \phi(X_t)$ be a learned representation. A representation break occurs when

$$P(Z_t \mid R_k) \neq P(Z_t \mid R_{k+1}).$$

A predictive break occurs when

$$P(Y_t \mid X_t, R_k) \neq P(Y_t \mid X_t, R_{k+1}).$$

A loss break occurs when

$$\mathbb{E}[L(Y_t, \hat{f}(X_t)) \mid R_k] \neq \mathbb{E}[L(Y_t, \hat{f}(X_t)) \mid R_{k+1}].$$

Define model decay in regime k as

$$MD_k = \mathbb{E}[L(Y_t, \hat{f}_{old}(X_t)) \mid R_k] - \mathbb{E}[L(Y_t, \hat{f}_k(X_t)) \mid R_k].$$

A high MD_k means the old model has lost structural relevance.

27 Strategic and Adversarial Breaks

In adversarial environments, breaks may be induced by strategic agents. Let a_t denote adversarial action. Then

$$X_t \sim P_{\theta_{\rho(t)}, a_t}.$$

The adversary may choose

$$a_t \in \arg \max_{a \in A} L(\theta, \delta(X_t^a)),$$

where δ is the statistician's decision rule.

A strategic break occurs when the adversary changes the data-generating mechanism:

$$a_k \neq a_{k+1}.$$

This is relevant for misinformation, sanctions evasion, financial fraud, cyber conflict, manipulated surveys, and wartime reporting.

28 Institutional and Measurement Breaks

Let Y_t^* be the latent variable of interest and Y_t the observed measurement. Suppose

$$Y_t = M_{\rho(t)}(Y_t^*) + u_t,$$

where M_k is the measurement system in regime k .

A measurement break occurs when

$$M_k \neq M_{k+1}.$$

Examples include:

- changes in GDP accounting;
- new inflation baskets;
- altered unemployment definitions;
- wartime censorship;
- new crime-reporting rules;
- platform moderation changes;
- revised casualty-reporting standards.

29 Hypothesis Testing

The fundamental null hypothesis is not “no effect.” It is “no break”:

$$H_0 : K = 1.$$

The alternative is

$$H_1 : K > 1.$$

For regression:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_K.$$

For causal effects:

$$H_0 : ATE_1 = ATE_2 = \cdots = ATE_K.$$

For causal graphs:

$$H_0 : G_1 = G_2 = \cdots = G_K.$$

For measurement:

$$H_0 : M_1 = M_2 = \cdots = M_K.$$

30 Decision Theory

Let a be an action and $L(a, \theta_k)$ be the loss in regime k . A global decision rule based on regime uncertainty minimizes

$$a^* = \arg \min_a \sum_{k=1}^K \mathbb{P}(R_T = k \mid \mathcal{D}_T) \mathbb{E}[L(a, \theta_k) \mid R_T = k, \mathcal{D}_T].$$

If future break risk matters, the decision rule becomes

$$a^* = \arg \min_a [(1 - BR_h) \mathbb{E}[L(a, \theta_T) \mid R_T] + BR_h \mathbb{E}[L(a, \theta_{\text{new}}) \mid R_{\text{new}}]].$$

This applies to monetary policy, military mobilization, portfolio risk, sanctions design, crisis response, and machine-learning deployment.

31 Simulation Design

A simple simulation design for structural-break-first analysis is:

$$X_t = \begin{cases} \mu_1 + \sigma_1 \varepsilon_t, & 1 \leq t \leq \tau_1, \\ \mu_2 + \sigma_2 \varepsilon_t, & \tau_1 < t \leq \tau_2, \\ \mu_3 + \sigma_3 \varepsilon_t, & \tau_2 < t \leq T. \end{cases}$$

With

$$\varepsilon_t \sim N(0, 1),$$

the likelihood cost for a segment I_k is

$$C(I_k) = \frac{n_k}{2} \log(\hat{\sigma}_k^2) + \frac{n_k}{2}.$$

The simulation may compare:

- pooled estimation;
- known-break estimation;
- estimated-break estimation;
- misspecified no-break forecasting;
- online break-risk forecasting.

32 Reporting Standard

A structural-break-first statistical report should include:

1. estimated number of regimes:

$$\widehat{K};$$

2. estimated break field:

$$\widehat{\mathcal{B}} = \{\widehat{\tau}_1, \dots, \widehat{\tau}_{\widehat{K}-1}\};$$

3. break uncertainty intervals:

$$[\tau_{k,L}, \tau_{k,U}];$$

4. regime-local parameters:

$$\widehat{\theta}_1, \dots, \widehat{\theta}_{\widehat{K}};$$

5. break-type classification;

6. Structural Break Index:

$$SBI;$$

7. horizon-specific break risk:

$$BR_h;$$

8. warning against global inference when structural instability is high.

33 Comparative Framework

Table 6: Classical versus structural-break-first inference

Dimension	Classical orientation	Structural-break-first orientation
Primitive object	Parameter	Regime structure
Default assumption	Stability	Local stability
Breaks	Exceptions	Central objects
Global mean	Primary	Derived
Causal effect	Often pooled	Regime-local
Forecast risk	Noise-driven	Noise plus break risk
Model failure	Residual anomaly	Possible new regime
Policy inference	Universal coefficient	Regime-contingent rule

34 Limitations

The framework has limitations.

First, excessive segmentation can overfit noise. This requires penalties, minimum segment lengths, and out-of-sample validation.

Second, break detection can be uncertain when transitions are gradual rather than abrupt.

Third, regime interpretation requires substantive knowledge. A statistical break is not automatically an economic, political, military, or institutional break.

Fourth, causal identification across regimes is difficult when treatment assignment, measurement, and institutional context all change simultaneously.

Fifth, ontological breaks may invalidate numerical comparison across regimes.

35 Conclusion

The structural-break-first statistical framework treats instability as fundamental. It replaces the single global data-generating process with a sequence of locally stable regimes separated by structural breaks. It thereby changes the order of inference:

classical: estimate parameter, then test stability;

structural-break-first: estimate stability, then estimate parameter.

The framework is especially useful in domains where crises, wars, institutions, markets, strategic behavior, and technological shifts alter the statistical meaning of data. Its central doctrine is:

No global inference without structural stability.

References

- [1] Bai, J. (1997). Estimating multiple breaks one at a time. *Econometric Theory*, 13, 315–352.
- [2] Bai, J. and Perron, P. (1998). Estimating and testing linear models with multiple structural changes. *Econometrica*, 66, 47–78.
- [3] Box, G. E. P. and Jenkins, G. M. (1970). *Time Series Analysis: Forecasting and Control*. Holden-Day.
- [4] Chow, G. C. (1960). Tests of equality between sets of coefficients in two linear regressions. *Econometrica*, 28, 591–605.
- [5] Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press.
- [6] Hastie, T., Tibshirani, R., and Friedman, J. (2009). *The Elements of Statistical Learning*. Springer.
- [7] Hayek, F. A. (1945). The use of knowledge in society. *American Economic Review*, 35, 519–530.
- [8] Keynes, J. M. (1936). *The General Theory of Employment, Interest and Money*. Macmillan.
- [9] Lucas, R. E. (1976). Econometric policy evaluation: A critique. In *The Phillips Curve and Labor Markets*. Carnegie-Rochester Conference Series.
- [10] Merton, R. C. (1973). An intertemporal capital asset pricing model. *Econometrica*, 41, 867–887.
- [11] Pearl, J. (2009). *Causality*. Cambridge University Press.
- [12] Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66, 688–701.
- [13] Schelling, T. C. (1960). *The Strategy of Conflict*. Harvard University Press.
- [14] Tukey, J. W. (1977). *Exploratory Data Analysis*. Addison-Wesley.
- [15] Waltz, K. N. (1979). *Theory of International Politics*. Addison-Wesley.

Glossary

Break Field

The set of estimated or true structural break dates,

$$\mathcal{B} = \{\tau_1, \dots, \tau_{K-1}\}.$$

Break Risk

The probability that the current regime will fail or change before a specified horizon:

$$BR_h = \mathbb{P}(\tau_{\text{next}} \leq T + h \mid \mathcal{D}_T).$$

Causal Break

A structural break in which the causal graph or causal mechanism changes across regimes.

Concept Drift

A machine-learning phenomenon in which the conditional relationship between predictors and outcomes changes over time.

Geopolitical Break

A structural change in the international system, such as a shift in alliances, sanctions, deterrence, polarity, or conflict structure.

Global Inference

Inference using the entire sample as if it were generated by one stable data-generating process.

Institutional Break

A change in rules, laws, norms, bureaucracies, incentives, or reporting systems that alters the data-generating process.

Local Stability

Stability that holds inside a regime but not necessarily across the entire sample.

Measurement Break

A change in the observation or reporting process, such that the same latent variable is measured differently across regimes.

Ontological Break

A break in which the meaning, support, or identity of the statistical object changes.

Regime

A locally stable statistical environment characterized by an interval, a probability law, and a parameter.

Regime Map

The function $\rho(t)$ assigning each observation to a regime.

Regime-Local Parameter

A parameter estimated within a single regime rather than across the full sample.

Strategic Break

A break induced by purposeful agents who alter behavior, reporting, manipulation, or concealment.

Structural Break

A discontinuity in the data-generating process.

Structural Break Index

The share of total variance attributable to between-regime differences:

$$SBI = \frac{\sum_{k=1}^K w_k (\mu_k - \bar{\mu})^2}{\text{Var}(X)}.$$

Structural Dispersion

The component of variance generated by differences across regime means:

$$SD = \sum_{k=1}^K w_k (\mu_k - \bar{\mu})^2.$$

Structural-Break-First Inference

An inferential doctrine in which the regime structure is estimated before global parameters are interpreted.

Warfare Economic Break

A structural change caused by mobilization, sanctions, blockade, destruction, logistical disruption, or military production reallocation.

The End